

Graßmann, Geometry and Mechanics

Graßmann Bicentennial Conference
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Introduction

- Many of the capabilities of Graßmann's algebra are still not well-enough known.

This paper briefly draws attention to just one of these:

- The ability to distinguish *mathematically* between points and vectors.
- We will look at just two applications of this capability:
 - Graßmannian geometry.
 - Graßmannian mechanics.

None of the examples will require a metric.

- The ability to distinguish between points and vectors appears to have been ignored ever since the adoption of Gibbs three-dimensional vector algebra as the dominant vectorial paradigm.

Connection to the Sesquicentennial Conference

In the Sesquicentennial Conference Professor Dr Alvin Swimmer presented a paper *The completion of Graßmann's Natur-Wissenschaftliche Methode*, in which he discussed the distinction between points and vectors, and applied the results to the expression of Newton's Second Law.

The paper to be presented today bears a similar message as that most important contribution: that many of the capabilities of Graßmann's algebra are still not well-enough known.

Today's talk will take a somewhat different approach to that adopted by Professor Swimmer. But I hope to arrive at the same conclusions.

Overview

- Graßmann's contribution and Gibbs' influence
- Geometry
 - *The exterior product*
 - Points and vectors
 - Bivectors and bound vectors
 - *The regressive product*
 - Geometric constructions
 - Geometric formulas
 - Geometric theorems

Mechanics

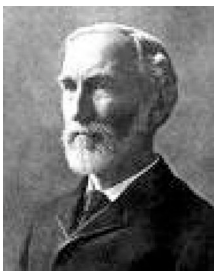
- Force
- Momentum
- Newton's Second Law

Graßmann



- The capability of Graßmann's algebra to allow for any number of dimensions is the key to distinguishing points from vectors.
 - It can easily use a 4-dimensional linear space to create an algebra of points and vectors in a space of three vector dimensions.
- The algebras of Hamilton and Gibbs could not do that as they were focused on three dimensions only.

Gibbs



- Gibbs three-dimensional vector algebra lost the point completely because it could not deal with the four dimensional linear space required.
 - Graßmann's algebra subsumes Gibbs' vector algebra, and it allows for points. Why not adopt the system that allows for both?
- Michael Crowe in his *A History of Vector Analysis* [1967, p159] comments:
 "Having noted that for geometrical applications, multiple algebra will generally take the form of a point or a vector analysis, Gibbs stated that
 ... in mechanics, kinematics, astronomy, physics or crystallography, Graßmann's point analysis will rarely be wanted."
 - In this talk we will agree with Gibbs that for geometrical applications a point *and* vector analysis is wanted. But for mechanics, we will disagree that it will rarely be wanted.

Graßmann algebra

- To construct a Graßmann algebra:
 - Take a linear space of dimension n with n basis elements.
 - Generate a new linear space by taking the *exterior product* of all these basis elements in all combinations, and keeping only those that are essentially different.
 This linear space is a Graßmann algebra.

The basis of a Grassmann algebra of 3-space is the collection of elements shown in the palette.

★ $\mathcal{B}_3$; BasisPalette

Basis Palette			
Λ_0	Λ_1	Λ_2	Λ_3
1	$\mathbf{e}_1$	$\mathbf{e}_1 \wedge \mathbf{e}_2$	$\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$
	$\mathbf{e}_2$	$\mathbf{e}_1 \wedge \mathbf{e}_3$	
	$\mathbf{e}_3$	$\mathbf{e}_2 \wedge \mathbf{e}_3$	

- At this stage, we do not call the elements $\mathbf{e}_i$ vectors, because geometrically *a vector is only one possible interpretation*.

Vector spaces

- An n -dimensional *vector* space needs n elements *interpreted as vectors* in its basis.

$$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n\}$$

- Vectors have *direction* but *no position*.
So they are somewhat difficult to depict ...

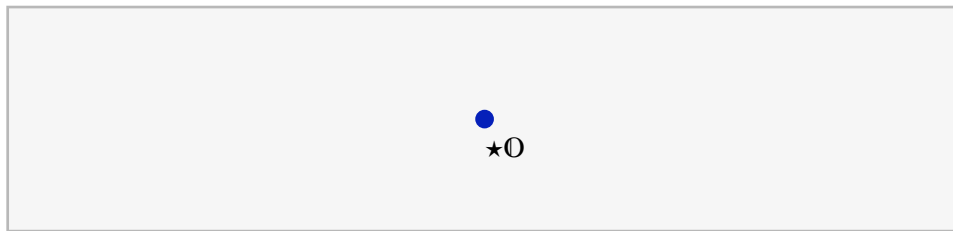


One way to depict an unlocated entity is to show it in *many* locations.

Bound vector spaces

- To describe *points* we need to add an *extra* element to the basis *interpreted* as an *origin point*.

$$\{\star 0, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n\}$$



In order to describe position it is necessary to have a reference point.

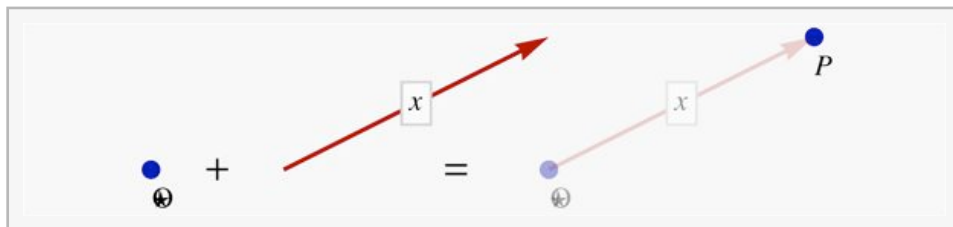
- We will call this new space a *bound vector space*.
- The Grassmann algebra of a *bound* 3-dimensional vector space has 2^4 basis elements.

Basis Palette				
Λ_0	Λ_1	Λ_2	Λ_3	Λ_4
1	$\star 0$	$\star 0 \wedge e_1$	$\star 0 \wedge e_1 \wedge e_2$	$\star 0 \wedge e_1 \wedge e_2 \wedge e_3$
	e_1	$\star 0 \wedge e_2$	$\star 0 \wedge e_1 \wedge e_3$	
	e_2	$\star 0 \wedge e_3$	$\star 0 \wedge e_2 \wedge e_3$	
	e_3	$e_1 \wedge e_2$	$e_1 \wedge e_2 \wedge e_3$	
		$e_1 \wedge e_3$		
		$e_2 \wedge e_3$		

Points

- A *point* is the sum of the origin and a vector.

$$\mathbf{P} = \star 0 + \mathbf{x}$$



The sum of the origin and a vector is a point.

- The vector $\mathbf{x}$ is called the *position vector* of the point $\mathbf{P}$.
- The *sum* of a point and a vector is another point.

$$(\star 0 + \mathbf{x}) + \mathbf{y} = \star 0 + (\mathbf{x} + \mathbf{y})$$

The *difference* between two points is a vector because the origin cancels.

$$(\star 0 + \mathbf{x}) - (\star 0 + \mathbf{y}) = \mathbf{x} - \mathbf{y}$$

- Vectors have direction but no position.
Points have position but no direction.

Weighted points

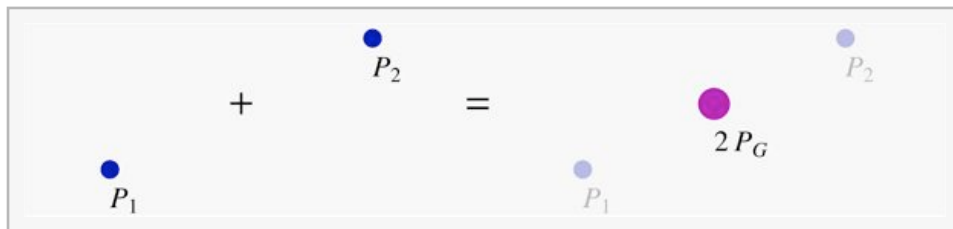
- To ensure an algebra involving points is a linear algebra, we must allow scalar multiples of points.
- A scalar multiple of a point is called a *weighted point*.

$$\mathbf{w} \mathbf{P} = \mathbf{w} (\star 0 + \mathbf{x})$$

- The geometric *interpretation* of a weighted point is as a *point with a weight attached*.

Sums of weighted points

- A sum of two points is a point located mid-way between them, but with a weight of 2 attached.



The sum of two points is a point of weight 2 located mid-way between them

$$\mathbf{P}_1 + \mathbf{P}_2 = (\star 0 + \mathbf{x}) + (\star 0 + \mathbf{y}) = 2 \left(\star 0 + \frac{\mathbf{x} + \mathbf{y}}{2} \right) = 2 \mathbf{P}_G$$

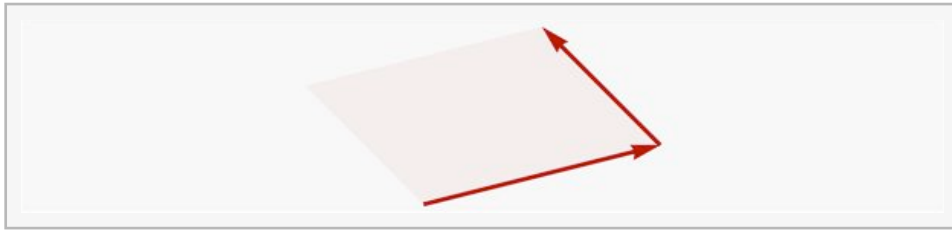
- Weighted points can model point masses.
- A sum of point masses gives a point mass whose
 - mass is equal to the total mass.
 - position is equal to the centre of mass.
- This is the type of property we need for modelling systems of particles.

Bivectors and *m*-vectors

- The exterior product of two vectors $\mathbf{x}$ and $\mathbf{y}$ is called a *simple bivector*, or *2-blade*.

$$\mathbf{B} = \mathbf{x} \wedge \mathbf{y}$$

- Bivectors (like vectors) are located *nowhere*.



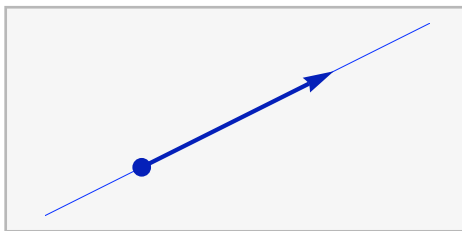
Depicting a simple bivector. This bivector is located nowhere.

- A sum of simple bivectors is also called a bivector (but may not be simple).
- The exterior product of m vectors is called a *simple m -vector*, or *m -blade*.
- Bivectors are precisely the type of entity we require to model *moments* in mechanics.

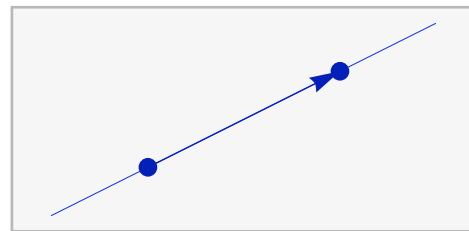
Bound vectors and bound m -vectors

- The exterior product of a point and a vector, or of two points, is a *bound vector*.

$$\mathbb{B} \equiv \mathbf{P} \wedge \mathbf{x} \equiv \mathbf{P} \wedge (\mathbf{Q} - \mathbf{P}) \equiv \mathbf{P} \wedge \mathbf{Q}$$



Point $\wedge$ vector depiction



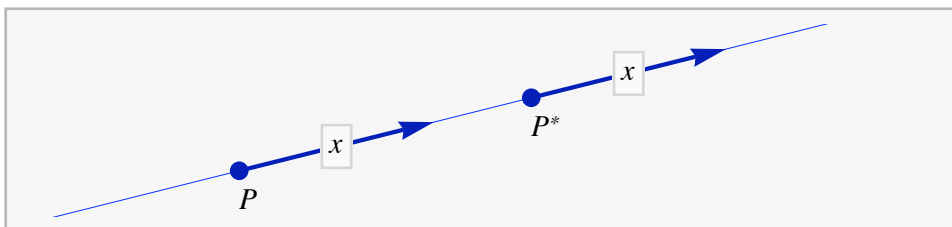
Point $\wedge$ point depiction

Two different depictions of a bound vector in its line.

- A sum of bound vectors may not be reducible to a bound vector.
- The exterior product of a point and m other points or vectors, is a *bound m -vector*.

Bound vectors can model lines and forces

- Bound vectors are located *anywhere* in their line.



A bound vector displaying its ability to slide along its line.

$$(\mathbf{P} - \mathbf{P}^*) \wedge \mathbf{x} \equiv \mathbf{0} \quad \Rightarrow \quad \mathbf{P} \wedge \mathbf{x} \equiv \mathbf{P}^* \wedge \mathbf{x}$$

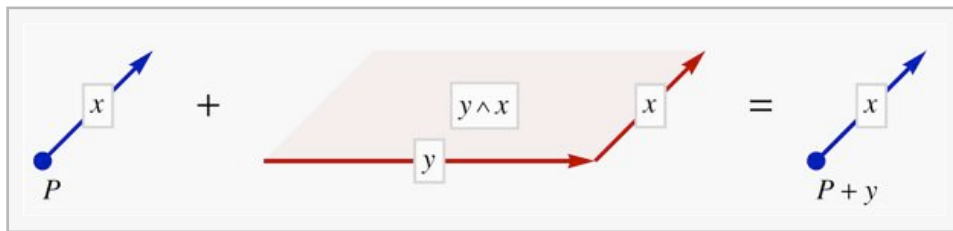
All textbooks tell us that a physical force has a *line of action*, and that a *sum of forces may not be reducible to a single force*.

Bound vectors are precisely the type of entity we need for modelling lines and forces.

- Bound m -vectors are located anywhere in their m -plane.
They are precisely the type of entity we need for modelling m -planes in Grassmannian geometry.

The sum of a bound vector and a bivector

- The sum of a bound vector and a bivector which have a vector in common is a parallel bound vector.



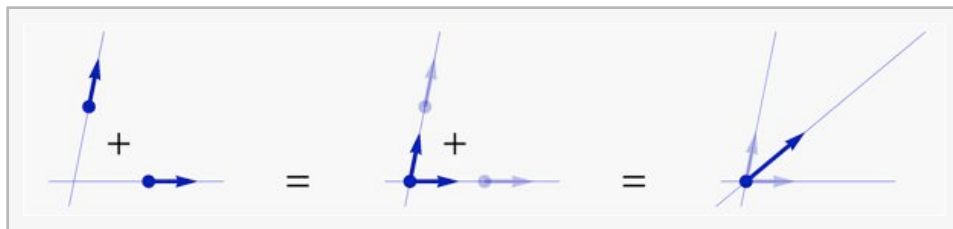
The bivector as a carrier of bound vectors.

$$\mathbf{P} \wedge \mathbf{x} + \mathbf{y} \wedge \mathbf{x} = (\mathbf{P} + \mathbf{y}) \wedge \mathbf{x}$$

- A vector can be considered as a *carrier* of points.
A bivector can be considered as a *carrier* of bound vectors.

The sum of two intersecting bound vectors

- The sum of two intersecting bound vectors is a bound vector through their point of intersection.



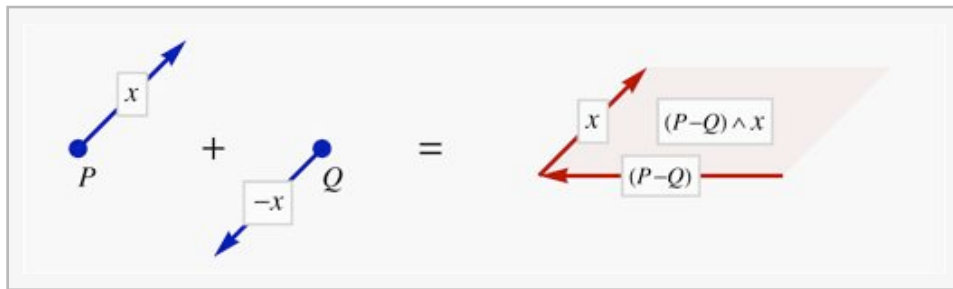
The sum of two non-parallel bound vectors in the plane is a bound vector through their point of intersection.

$$\mathbf{P} \wedge \mathbf{x} + \mathbf{Q} \wedge \mathbf{y} = \mathbf{R} \wedge \mathbf{x} + \mathbf{R} \wedge \mathbf{y} = \mathbf{R} \wedge (\mathbf{x} + \mathbf{y})$$

- In mechanics this translates to:
Forces acting through the same point can be resolved to a *single* force.

The sum of two opposite bound vectors

- The sum of two oppositely-sensed bound vectors is a bivector.



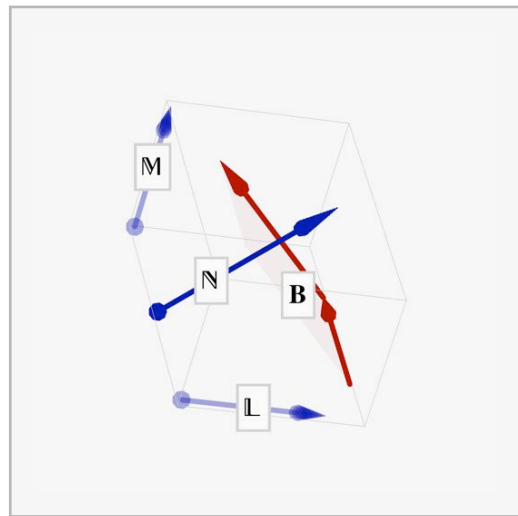
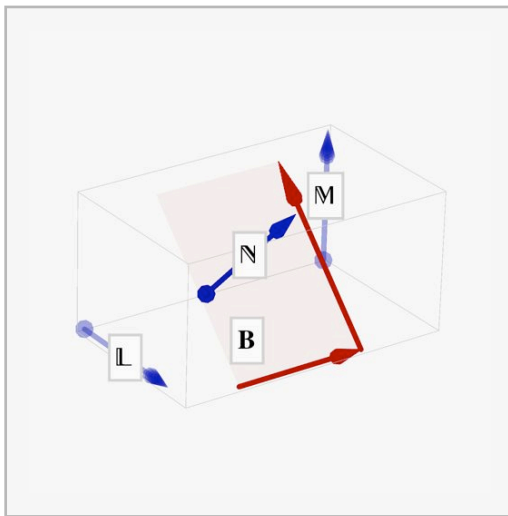
The sum of two bound vectors, whose vectors are equal but of opposite sense, is a bivector.

$$\mathbf{P} \wedge \mathbf{x} + \mathbf{Q} \wedge (-\mathbf{x}) = (\mathbf{P} - \mathbf{Q}) \wedge \mathbf{x}$$

- Thus, the sum of these two bound entities (bound to their lines) reduces to a free entity (located nowhere).
- In mechanics this translates to:
Two oppositely sensed forces generate a *couple*.

The sum of two bound vectors

- The sum of two bound vectors $\mathbb{L}$ and $\mathbb{M}$ may always be reduced to the sum of a bound vector $\mathbb{N}$ and a bivector $\mathbf{B}$ (either may be zero).



The sum of two bound vectors. View 1. The sum of two bound vectors. View 2.

$$\mathbb{L} + \mathbb{M} = \mathbb{N} + \mathbf{B}$$

- In mechanics this translates to:
The sum of two forces in space is not generally reducible to a single force.

The sum of bound vectors

- A sum of bound vectors may always be reduced to the sum of:
 - a bound vector $\mathbf{P} \wedge \mathbf{X}$ through an arbitrarily chosen point $\mathbf{P}$
 - a bivector $\mathbf{B}_P$ (either of which may be zero).

$$\sum_i \mathbf{P}_i \wedge \mathbf{x}_i = \mathbf{P} \wedge \sum_i \mathbf{x}_i + \sum_i (\mathbf{P}_i - \mathbf{P}) \wedge \mathbf{x}_i = \mathbf{P} \wedge \mathbf{X} + \mathbf{B}_P$$

Sum of bound vectors $=$ Bound vector $+$ Bivector

- This reduction trick shows clearly how the bivector $\mathbf{B}_P$ can be dependent on a point $\mathbf{P}$, and yet be a free entity (located nowhere).

The regressive product

- In order to do geometry with the algebraic entities we can now construct, we need a way to find intersections, or *common factors*.

The key is Grassmann's *regressive product*.

- In his *Ausdehnungslehre* of 1862 Section 95 (translated by Lloyd C. Kannenberg) Grassmann states (my parentheses):

"If q and r are the orders (grades) of two magnitudes A and B , and n that of the principal domain (base linear space), then the order (grade) of the product $[A B]$ is *first* equal to $q+r$ if $q+r$ is smaller than n , and *second* equal to $q+r-n$ if $q+r$ is greater than or equal to n ."

In terms of modern notation for the exterior and regressive products:

$$\begin{aligned} [A B] &\Leftrightarrow A \wedge B & q + r < n & \quad (\text{exterior product}) \\ [A B] &\Leftrightarrow A \vee B & q + r \geq n & \quad (\text{regressive product}) \end{aligned}$$

- Thus Grassmann's $[A B]$ product was actually two products, the regressive product was hidden in the exterior product!

This was a beautiful and powerful notational device, but leaving the product type to be chosen by the grades of its factors, rather than by being explicitly denoted, may well be the reason why Grassmann's regressive product is still not well known today.

Projective geometry

- A Grassmann algebra on a bound vector space equipped with both the exterior and regressive products, is now all that we need to explore projective geometry.

Projective Geometry	Grassmann Geometry	Grade
Point	Point	1
Point at infinity	Vector	1
Line	Bound vector	2
Line at infinity	Bivector	2
Plane	Bound bivector	3
Plane at infinity	Trivector	3
Multiplane	Bound $(m-1)$ -vector	m
Multiplane at infinity	m -vector	m

- The key is to *identify m -planes at infinity with $(m+1)$ -vectors*. For example

point at infinity $\iff$ vector
 line at infinity $\iff$ bivector
 plane at infinity $\iff$ trivector

This is because the regressive product (intersection) of parallel bound elements is their common vectorial direction.

Geometric constructions

- Exploring Grassmann non-metric geometry often simply involves expanding and simplifying geometric formulas composed of products (exterior and regressive) of points, lines (bound vectors), planes (bound bivectors), ...
- For example, Grassmann showed that some geometric formulas can be viewed as *a prescription to construct*.

Here is a recipe for constructing a conic in the plane.

1. Construct the fixed points and lines: $\mathbf{P}_0, \mathbf{P}_1, \mathbf{P}_2, \mathbf{L}_1$, and $\mathbf{L}_2$.
2. Construct a line $\mathbf{L}_c$ through $\mathbf{P}_2$ to intersect line $\mathbf{L}_2$ in point $\mathbf{Q}_2$.
3. Construct a line $\mathbf{L}_b$ through $\mathbf{Q}_2$ and $\mathbf{P}_0$ to intersect line $\mathbf{L}_1$ in point $\mathbf{Q}_1$.
4. Construct a line $\mathbf{L}_a$ through $\mathbf{Q}_1$ and $\mathbf{P}_1$ to intersect line $\mathbf{L}_c$ in point $\mathbf{P}$.
5. Repeat the construction by drawing a new line $\mathbf{L}_c$ through $\mathbf{P}_2$.

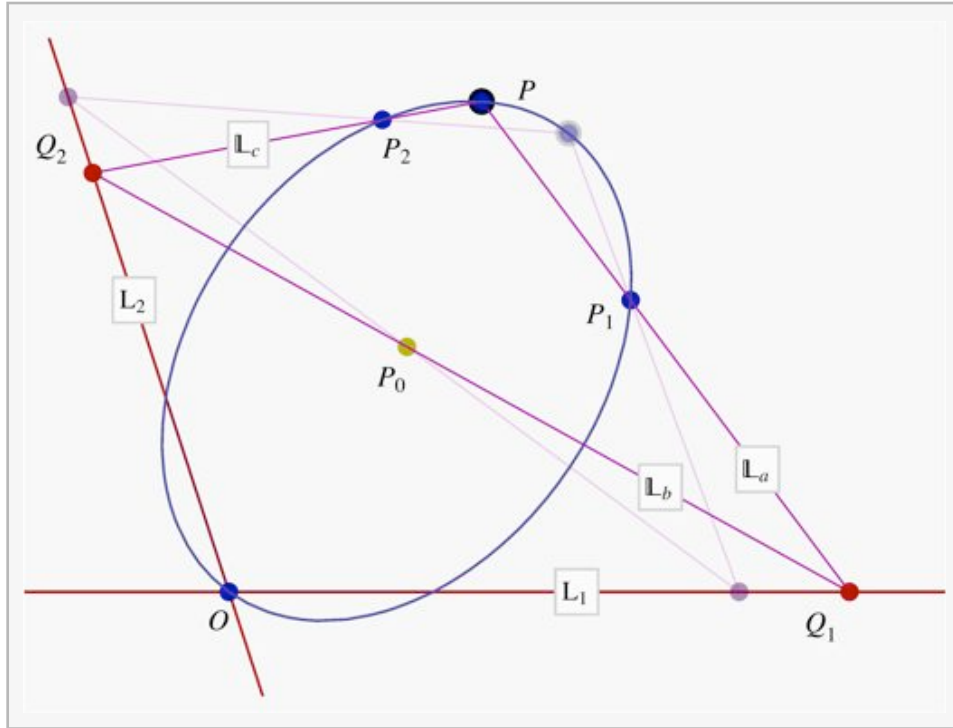
And here is Grassmann's formula encoding this construction.

$\mathbf{P}$ is a variable point
 $\mathbf{P}_0, \mathbf{P}_1, \mathbf{P}_2$ are fixed points
 $\mathbf{L}_1, \mathbf{L}_2$ are fixed lines

$$\mathbf{P} \wedge \mathbf{P}_1 \wedge (\mathbf{L}_1 \vee (\mathbf{P}_0 \wedge (\mathbf{L}_2 \vee (\mathbf{P} \wedge \mathbf{P}_2)))) = 0$$

The variable point $\mathbf{P}$ occurs twice, leading to a curve of the second degree.

This equation *is* the equation of the conic. Substituting symbolic or numeric coordinates reduces it to the expected algebraic relation from which the ellipse below is plotted.



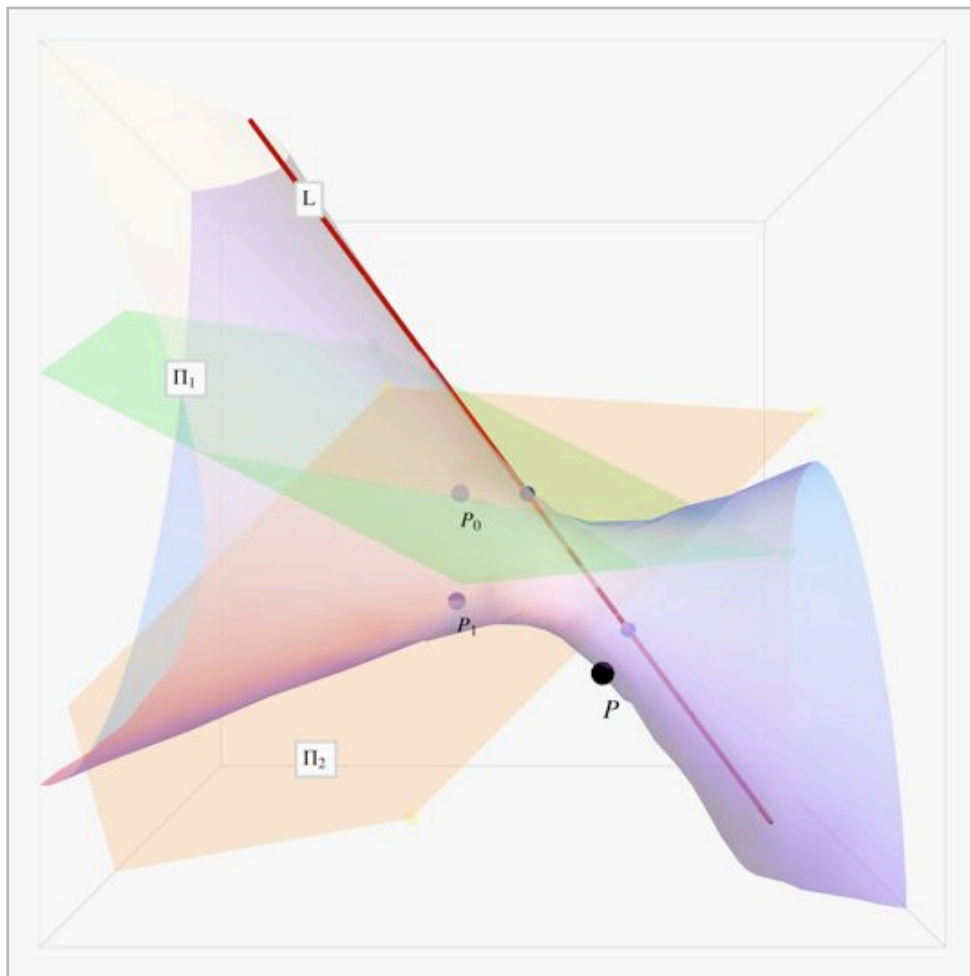
Constructing a conic section through 5 points

Geometric formulas in higher dimensions

- There is no restriction on composing geometric formulas in higher dimensions. Here is a 3-dimensional version of the planar conic.

$$\mathbf{P} \wedge \mathbf{P}_1 \wedge (\Pi_1 \vee (\mathbf{P}_0 \wedge (\Pi_2 \vee (\mathbf{P} \wedge \mathbf{L})))) = 0$$

- $\mathbf{P}$ is a variable point
- $\mathbf{P}_0, \mathbf{P}_1$ are fixed points
- $\mathbf{L}$ is a fixed line
- Π_1, Π_2 are fixed planes



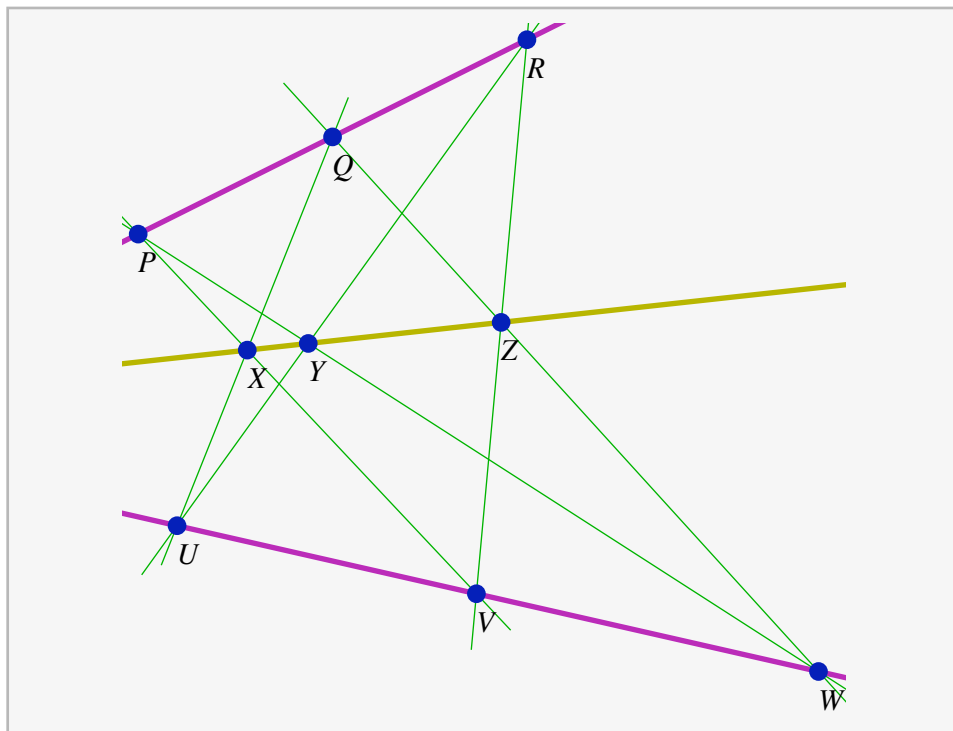
Constructing a conic from two points, a line and two planes

We can see the actual expressions used for the various geometric entities by hovering the mouse over them in the graphic.

Geometric theorems

- Any theorem that can be reduced to a geometric formula can be proved by adopting symbolic coordinates for the geometric entities, and simplifying the formula. For example, here is Pappus' Theorem:

Given two sets of collinear points P, Q, R , and U, V, W , then the intersection points X, Y, Z of line pairs PV and QU , PW and RU , QW and RV , are collinear.



Pappas' theorem.

- To prove the theorem using the Graßmannian approach, one only has to choose symbolic values for the points in an arbitrary basis, and compute the collinearity condition $\mathbf{X} \wedge \mathbf{Y} \wedge \mathbf{Z} == \mathbf{0}$.

$$\begin{aligned}
 \mathbf{X} &== (\mathbf{P} \wedge \mathbf{V}) \vee (\mathbf{Q} \wedge \mathbf{U}) \\
 \mathbf{Y} &== (\mathbf{P} \wedge \mathbf{W}) \vee (\mathbf{R} \wedge \mathbf{U}) \\
 \mathbf{Z} &== (\mathbf{Q} \wedge \mathbf{W}) \vee (\mathbf{R} \wedge \mathbf{V})
 \end{aligned}
 \qquad
 \mathbf{X} \wedge \mathbf{Y} \wedge \mathbf{Z} == \mathbf{0}$$

Proof of Pappus' theorem

We begin by declaring that we wish to work in the plane, and then define the points P, Q, R, and U, V, W by means of two vectors $\mathbf{u}$ and $\mathbf{v}$ and four scalars $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$ in such a way as to ensure the triplets of points are collinear.

```

★P2;
P = ★0 + v;
Q = ★0 + a v;
R = ★0 + b v;
U = ★0 + u;
V = ★0 + c u;
W = ★0 + d u;

```

The next stage is to compute the intersection points X, Y, and Z. We do this by using the *GrassmannAlgebra* functions `ToCommonFactor` and `ToPointForm`.

$$\mathbf{X} = \text{ToPointForm}[\text{ToCommonFactor}[(\mathbf{P} \wedge \mathbf{V}) \vee (\mathbf{Q} \wedge \mathbf{U})]]$$

$$\frac{(-1+a)cu}{-1+ac} + \frac{a(-1+c)v}{-1+ac} + \star 0$$

$$\mathbf{Y} = \text{ToPointForm}[\text{ToCommonFactor}[(\mathbf{P} \wedge \mathbf{W}) \vee (\mathbf{R} \wedge \mathbf{U})]]$$

$$\frac{(-1+b)du}{-1+bd} + \frac{b(-1+d)v}{-1+bd} + \star 0$$

$$\mathbf{Z} = \text{ToPointForm}[\text{ToCommonFactor}[(\mathbf{Q} \wedge \mathbf{W}) \vee (\mathbf{R} \wedge \mathbf{V})]]$$

$$\frac{(a-b)cdu}{ac-bd} + \frac{ab(c-d)v}{ac-bd} + \star 0$$

Finally we simplify the exterior product of the three points. A value of zero proves that they are collinear, and hence proves the theorem.

$$\text{Simplify}[\text{GrassmannExpandAndSimplify}[\mathbf{X} \wedge \mathbf{Y} \wedge \mathbf{Z}]]$$

$$0$$

The sum of forces

- We will now look at the way bound vectors and bivectors can be used to model systems of forces. Our discussion follows exactly (*mutatis mutandis*) our discussion on sums of bound vectors.
- A force $\mathbf{F}$ may be faithfully represented by a *bound vector* as the exterior product of *any point* $\mathbf{P}$ on its line of action with its *force vector* $\mathbf{f}$.

$$\mathbf{F} == \mathbf{P} \wedge \mathbf{f}$$

- The sum of any number of forces may always be reduced to the sum of:
 - a resultant force $\mathbf{P} \wedge \mathbf{F}$ through an arbitrarily chosen point $\mathbf{P}$
 - a moment bivector $\mathbf{M}_P$ (either of which may be zero).

$$\sum_i \mathbf{P}_i \wedge \mathbf{f}_i == \mathbf{P} \wedge \sum_i \mathbf{f}_i + \sum_i (\mathbf{P}_i - \mathbf{P}) \wedge \mathbf{f}_i == \mathbf{P} \wedge \mathbf{F} + \mathbf{M}_P$$

$$\text{Sum of forces} == \text{Resultant force} + \text{Moment}$$

- This reduction trick shows clearly how the moment bivector $\mathbf{M}_P$ can be dependent on a point $\mathbf{P}$, and yet be a free entity (located nowhere).

Equilibrium

- In Grassmannian terms, the *single* condition for the equilibrium of a rigid body is

$$\mathbf{F} == \mathbf{P} \wedge \mathbf{F} + \mathbf{M}_P == \mathbf{0}$$

- The three-dimensional vector algebra approach requires *two* conditions
 - the sum of the force *vectors* is equal to zero
 - the sum of the moment *vectors* about any point is zero
- These are valid *only* in a 3-space with a metric, because the moment is defined as a cross product.
- The Graßmannian approach is valid in any dimension with or without a metric. This includes the plane.

The momentum of a particle

- Textbooks rarely mention that *the momentum of a particle has a line of action*, just like a force. One author who does emphasize this point is E. A Milne in his *Vectorial Mechanics* [1948], although he does not use a Graßmannian approach.
- The *velocity of a point* is the derivative of its position vector.

$$\mathbf{P}(\mathbf{t}) == \star \mathbf{0} + \mathbf{x}(\mathbf{t})$$

$$\overset{\circ}{\mathbf{P}} == \frac{d\mathbf{P}}{d\mathbf{t}} == \overset{\circ}{\mathbf{x}}$$

- The *momentum of a particle* is a bound vector through the particle.

$$\mathbf{L} == \mathbf{P} \wedge \left(m \overset{\circ}{\mathbf{P}} \right) == (m \mathbf{P}) \wedge \overset{\circ}{\mathbf{P}} == m \mathbf{P} \wedge \overset{\circ}{\mathbf{P}}$$

Here

$\mathbf{L}$	is the particle momentum
m	is the particle mass
$\mathbf{P}$	is the particle position point
$\overset{\circ}{\mathbf{P}}$	is the particle velocity
$m \mathbf{P}$	is the particle point mass
$m \overset{\circ}{\mathbf{P}}$	is the particle momentum vector

The momentum of a system

- The discussion below follows exactly (*mutatis mutandis*) our discussion on sums of bound vectors and sums of forces.
- The sum of any number of particle momenta may always be reduced to the sum of:
 - a resultant momentum $\mathbf{P} \wedge \mathbf{L}$ through an arbitrarily chosen point $\mathbf{P}$
 - an angular momentum bivector $\mathbf{H}_P$ (either of which may be zero).

$$\sum_i \mathbf{P}_i \wedge \mathbf{L}_i \quad == \quad \mathbf{P} \wedge \sum_i \mathbf{L}_i + \sum_i (\mathbf{P}_i - \mathbf{P}) \wedge \mathbf{L}_i \quad == \quad \mathbf{P} \wedge \mathbf{L} + \mathbf{H}_\mathbf{P}$$

Sum of momenta == Resultant momentum + Angular momentum

- We can choose $\mathbf{P}$ to be the centre of mass $\mathbf{P}_G$, so that the resultant momentum becomes the momentum of the mass centre.

$$\mathbf{P} \wedge \mathbf{L} + \mathbf{H}_\mathbf{P} \quad == \quad \mathbf{P}_G \wedge \left(\mathbf{M} \overset{\circ}{\mathbf{P}}_G \right) + \mathbf{H}_{\mathbf{P}_G}$$

The rate of change of momentum

- The *momentum of the system* (referred to the centre of mass) is

$$\mathbf{L} \quad == \quad \mathbf{P}_G \wedge \left(\mathbf{M} \overset{\circ}{\mathbf{P}}_G \right) + \mathbf{H}_{\mathbf{P}_G}$$

- The *rate of change of momentum of the system* is then

$$\overset{\circ}{\mathbf{L}} \quad == \quad \overset{\circ}{\mathbf{P}}_G \wedge \left(\mathbf{M} \overset{\circ}{\mathbf{P}}_G \right) + \mathbf{P}_G \wedge \left(\mathbf{M} \overset{\circ\circ}{\mathbf{P}}_G \right) + \overset{\circ}{\mathbf{H}}_{\mathbf{P}_G}$$

Since the first term is zero because of the repeated factor $\overset{\circ}{\mathbf{P}}_G$, we obtain

$$\overset{\circ}{\mathbf{L}} \quad == \quad \mathbf{P}_G \wedge \left(\mathbf{M} \overset{\circ\circ}{\mathbf{P}}_G \right) + \overset{\circ}{\mathbf{H}}_{\mathbf{P}_G}$$

Thus the rate of change of momentum of a system of particles may be written as the sum of

- the rate of change of linear momentum of the total mass at the centre of mass
- the rate of change of angular momentum about the centre of mass.

Newton's Second Law

- In Grassmannian terms, Newton's Second Law can now be stated without reference to separate linear and angular components, since the bound vector / bivector representation includes both. The statement uses our new representation of forces and momenta as bound vectors.

The sum of forces acting on a system of particles or rigid bodies is equal to the time rate of change of momentum of the system.

- The Grassmannian version of Newton's Second Law is:

$$\mathbf{F} \quad == \quad \overset{\circ}{\mathbf{L}}$$

- We have seen that this can be written in bound vector / bivector terms as

$$\mathbf{P}_G \wedge \mathbf{F} + \mathbf{M}_{P_G} == \mathbf{P}_G \wedge \left(\overset{\circ\circ}{\mathbf{M}} \mathbf{P}_G \right) + \overset{\circ}{\mathbf{H}}_{P_G}$$

- This *one* equation can be seen to imply *both* the standard conditions:

$$\mathbf{F} == \overset{\circ\circ}{\mathbf{M}} \mathbf{P}_G$$

$$\mathbf{M}_{P_G} == \overset{\circ}{\mathbf{H}}_{P_G}$$

But the bivector form here allows validity in *any* number of dimensions.

- *Nowhere in this analysis has any metric or space dimension been required.*

Conclusions

- This paper has discussed just one of the capabilities of Grassmann's algebra which are still not well known:
 - The ability to distinguish between points and vectors.
- This appears to have been ignored due the dominance of Gibbs' three-dimensional vector algebra.

Today we have tried to show that, at least in geometry and mechanics, the distinction can lead straightforwardly to some powerful and insightful results.

- Grassmann's regressive product is also still not well known, perhaps as we saw, due to it having been hidden in the sophistication of his notation.

Acknowledgements

To Hermann Grassmann first and foremost in this bicentennial year I tender my most profound appreciation for his algebra.

To the conference organizers, and especially to Professor Dr. Petsche, my sincere thanks and appreciation for making this conference the enjoyable success that it is.

Reference

You can find a copy of this paper and other background material at

<http://sites.google.com/site/grassmannalgebra>